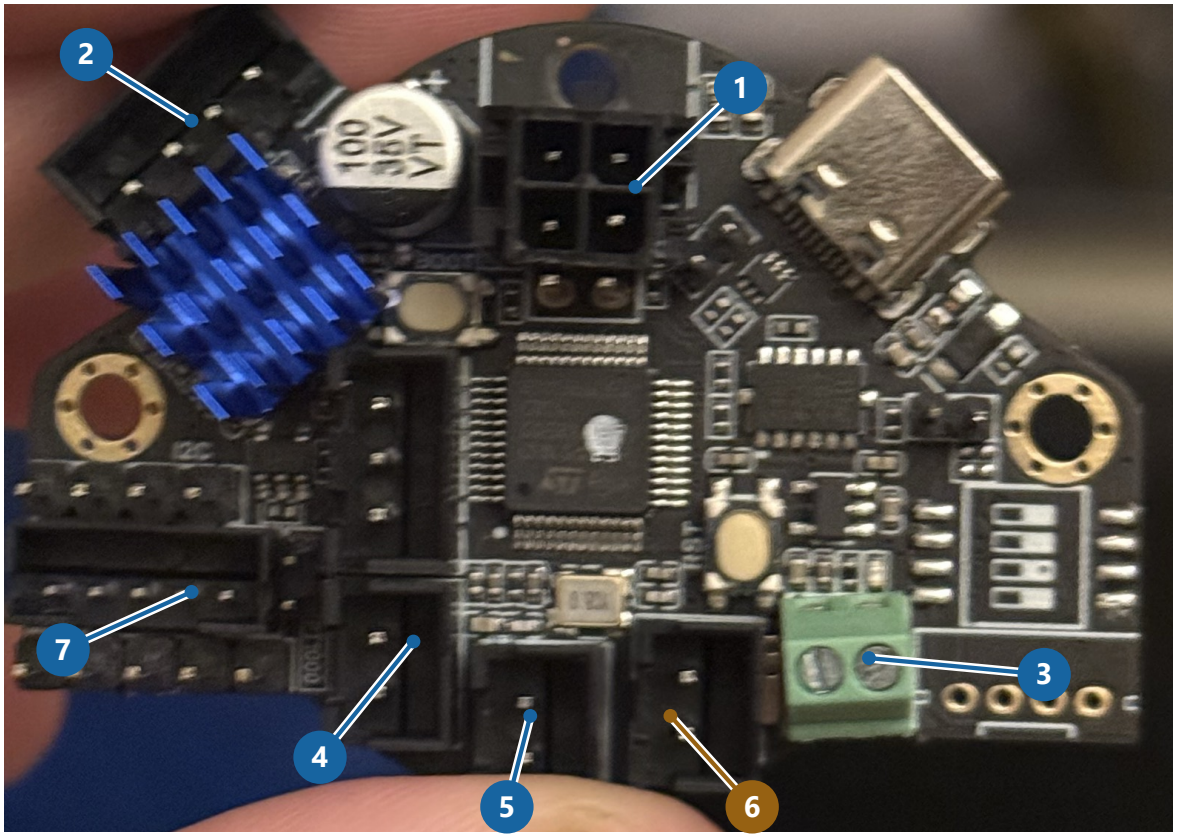
Your EBB36 toolhead map

Your photographed board / first-generation V1.1-V1.2 documentation / no MAX31865

Before assembly: unplug printer power and USB. Change plugs and jumpers only with both removed.



1. **Power + CAN**
   24 V, GND, CAN H, CAN L / page 5
2. **Extruder motor**
   E0 / EM four-contact plug / page 3
3. **X1C heater**
   Green E0 / HE0 screw pair / page 2
4. **X1C thermistor**
   TH0 / PA3 / page 2
5. **Two-wire hotend fan**
   PA0 / BTT FAN1 / page 2
6. **Four-wire blower**
   PA1 reserved; hold connection / page 6
7. **X + Y Hall sensors**
   Five-contact ENDSTOP / page 4

Assembly order

1 / Hotend leads

Heater, thermistor, hotend fan

2 / Motor + sensors

Measure coils; label Hall leads

3 / Main harness

24 V, GND and twisted CAN pair

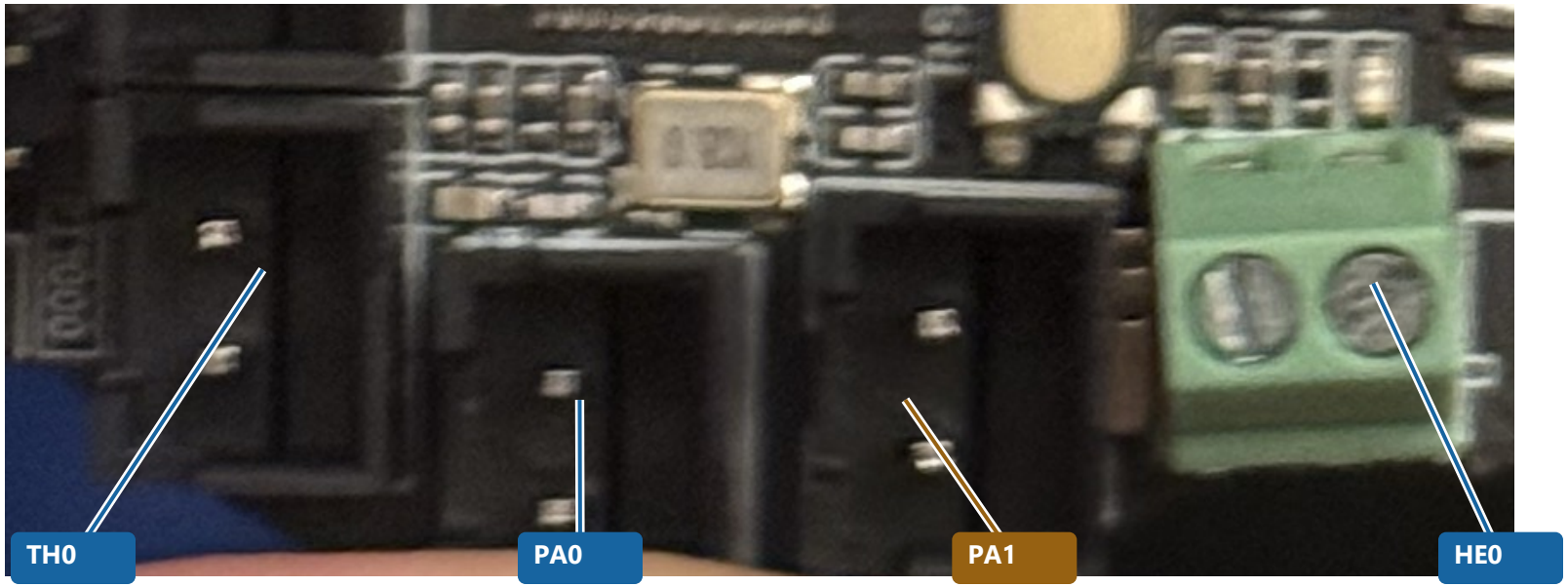
4 / Inspect + test

Jumpers, continuity, staged power

Wire the three hotend circuits

Bambu X1C heater + thermistor / Peerless DF3010B24H hotend fan, 24 V, 0.09 A

Board front, same orientation as page 1. Upper contacts point toward the MCU.



X1C heater

Green E0 / HE0 terminal
One heater lead in each clamp.
The heater pair has no polarity.

Control: **EBBCan:PB13**
Use a sound wire/ferrule termination; no solder-tinned strands in the clamp.

X1C thermistor

TH0, two-contact plug
Upper contact: **PA3**
Lower contact: **GND**

One thermistor lead per contact; no polarity.
PT1000 jumper stays **open** for the planned NTC connection.

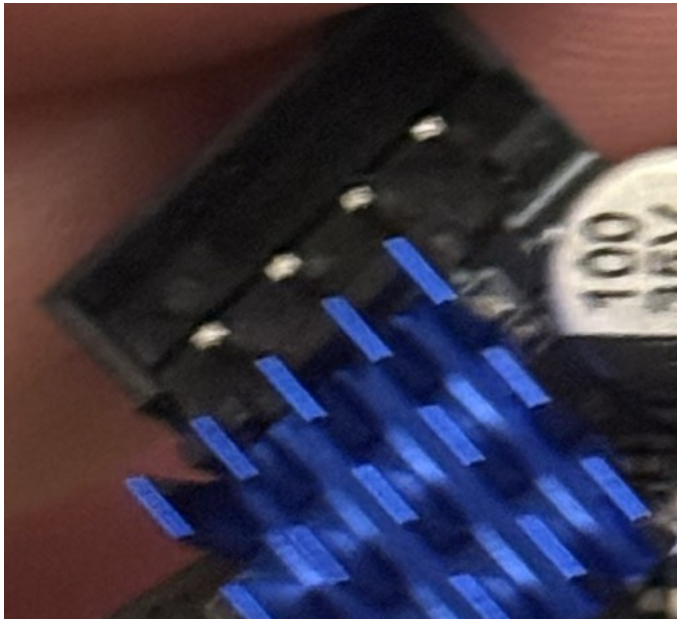
Hotend fan

PA0 / FAN1, two-contact plug
Upper: **PA0 / black fan negative**
Lower: **VIN / red fan positive**

This build uses PA0 for hotend cooling.
Configure it as the hotend fan; BTT's example assigns it differently.

Identify the extruder motor coils

Use the EBB36 onboard driver / measure with the motor unplugged from the board



E0 / EM motor connector

Use the angled four-contact socket at the upper left. Keep the existing motor lead if it is suitable.

Plug orientation and wire colors still need a continuity record. Do not guess from color.

BTT V1.2 schematic: P11 / E0

These are schematic pin numbers, not a cable-entry drawing.

P11 pin 1	**A2**	Coil A	Wire: _____
P11 pin 2	**A1**	Coil A	Wire: _____
P11 pin 3	**B1**	Coil B	Wire: _____
P11 pin 4	**B2**	Coil B	Wire: _____

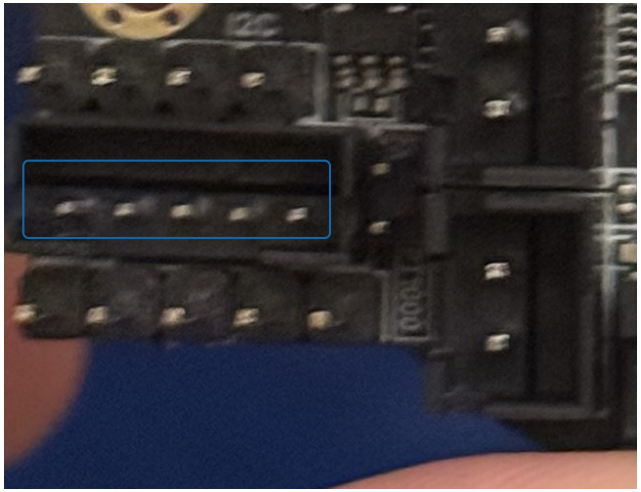
Measure, label, then crimp

1. Set the meter to resistance. Find two pairs with low, finite resistance; wires from different coils should read open.
2. Label one measured pair A and the other B. Each complete pair goes to its matching A or B contacts.
3. Identify actual connector pin 1 before repinning; then verify each wire through the finished lead. Never move a motor plug while powered.

Motor current, direction and rotation distance remain commissioning settings.

Connect X and Y Hall endstops

Five-contact ENDSTOP header / X signal = PB6 / Y signal = PB7 / PB5 unused



ENDSTOP is the middle row.

The four-contact I2C row is above it. Match the board-front orientation before inserting contacts.

Left to right in BTT's front view

+5 V	GND	PB6	PB5	PB7
Both VCC	Both GND	X output	Unused	Y output

The board already provides 10 kOhm pull-ups to 3.3 V on these inputs (BTT schematic R35-R37). The sensors use a 5 V supply while their open-collector outputs use 3.3 V signal pull-ups.

X sensor

VCC	———	+5 V
GND	———	GND
OUT	———	PB6

Y sensor

VCC	———	+5 V
GND	———	GND
OUT	———	PB7

Sensor leg reference
Original Allegro -UA package, branded face toward you, leads down:
1 = VCC, 2 = GND, 3 = OUT.

Confirm the actual part and orientation before soldering. Insulate each leg and support the lead bundle.

Build the power + CAN harness

Planned link: Raspberry Pi USB > Octopus Pro USB-to-CAN bridge > EBB36 CAN



Looking at the board front

GND

PSU V- / common ground

VIN

Protected +24 V

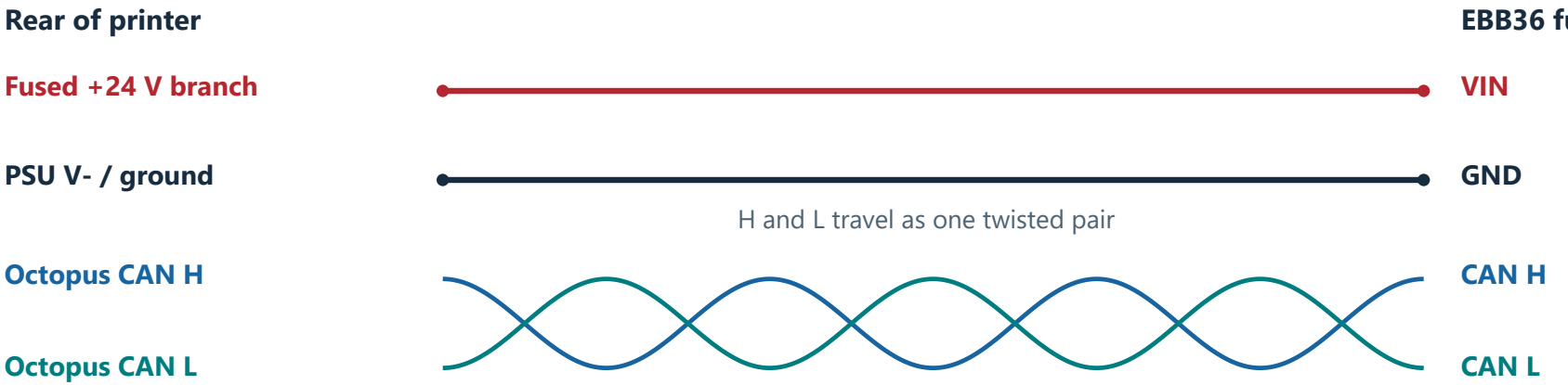
CAN L

Octopus CAN L

CAN H

Octopus CAN H

USB-C is at the upper right, as photographed. A plug's wire-entry view can be mirrored: verify every contact end to end instead of copying a left/right view.



Fit one 120 Ohm termination at each CAN end (Octopus and EBB). With all power removed and the bus connected, H-to-L should measure about 60 Ohm. Record cable colors and the selected branch fuse.

Prepare the four-wire blower separately

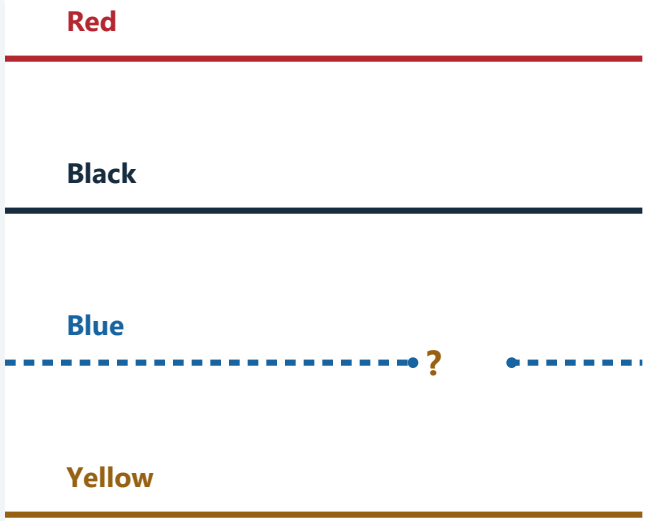
HXH HBB0524VD / 24 V / 0.25 A / control connection is not yet validated

Planning only: keep the blower unplugged while its PWM interface is checked. PA1 remains reserved.

STOCK BLOWER

Four leads have different jobs. Wire colors below come from the conversion reference, not an HXH datasheet.

Confirm the supplied fan's actual leads before terminating.



+24 V supply

Supply function only. Final branch and fan lead identification still need checking.

Permanent GND

Ground stays connected for the fan's separate logic control.

PWM control input

Logic voltage, drive circuit and control behavior need a verified specification or measurement.

Speed feedback / tach

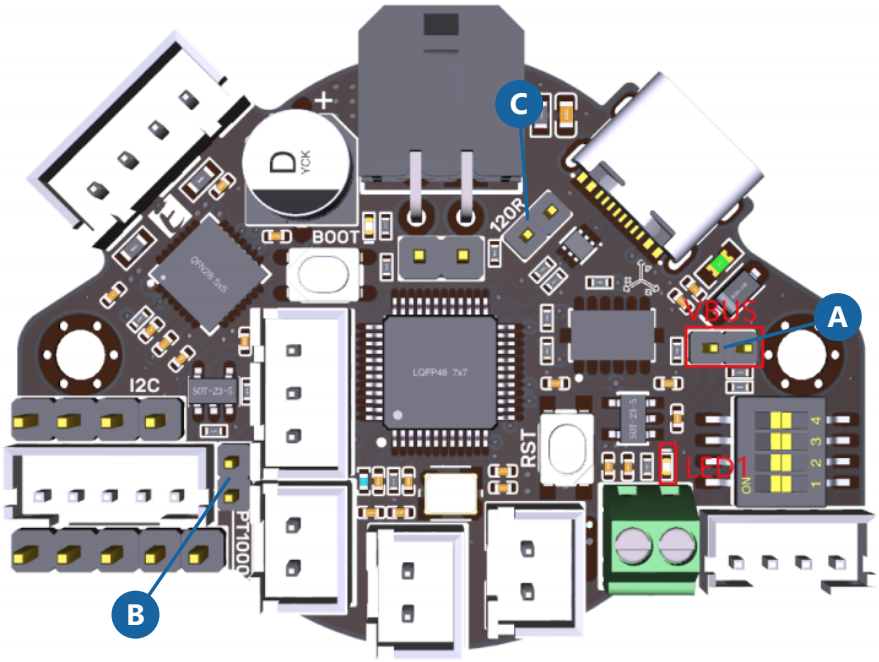
Leave unconnected and individually insulated if feedback is not used.

What BTT establishes
PA1 / FAN2 is a switched fan-power return. It is not a raw MCU logic pin. The fan header also carries VIN.

What is still needed
Establish a compatible PWM interface for this fan. Do not attach its PWM lead to 24 V or copy a control frequency without confirming it.

Set jumpers, inspect, then power in stages

Assembly is separate from flashing and heating / successful firmware flashes are not yet recorded



Normal 24 V + CAN operation

A VUSB: OPEN

Fit only for isolated USB-only electronics power.

B PT1000: OPEN

Planned NTC on TH0 uses the 4.7 kOhm pull-up.

C 120R: FIT

EBB is the CAN bus end. Octopus needs the other 120 Ohm termination.

Keep the PCB underside clear of the motor case and screws. Support the harness at the mount; leave a service loop clear of belts, fan blades and the nozzle.

1 / Check unpowered

Tug crimps; verify contact mapping, polarity, shorts and CAN termination. Select the fuse for the actual branch wiring and load.

2 / Bring up electronics

Disconnect heater and motor for flashing. Confirm both MCUs, CAN, room-temperature sensor reading, hotend fan and Hall states.

3 / Enable heat last

Power off before reconnecting loads. Confirm X1C sensor curve and heater checks, then test under supervision; finish blower and motion tests.

Primary sources (click labels; full source list accompanies this guide)

B1 BTT hardware | B2 V1.2 schematic | B3 V1.2 example | K1 CAN bus | K2 Multi-MCU homing | A1 Hall datasheet